



The AI Literacy Podcast

I. Task context, reflections, and legacy

Artificial Intelligence has transformed the world and reshaped the information age forever. Its power to generate outputs is impressing both novice and expert users due to the creative nature these algorithms exhibit. As audiovisual information becomes more accessible and faster to produce than ever before, algorithms that process images, audio, and text have become the core engines of this new revolution, with data considered not only the fuel but the gold of the century.

Mastering machine learning techniques can place you beyond simply being a knowledgeable developer. It gives you the ability to make AI more transparent by explaining to others how these algorithms operate, their implications, and how to consciously evaluate their applications.

Therefore, the idea of bringing together all the studied techniques to build an agent that participates in a podcast has the primary objective of exercising both dimensions: being a skilled Deep Learning programmer and caring about the people your work can reach.

Although content creation, audiovisual editing, and presentation skills are not part of the evaluated points, as you will see later in this document, we believe this opportunity is a good space to develop them. Furthermore, explaining your implementations and reasoning is an excellent reinforcement mechanism to better absorb and reflect on the learned content.

We also expect that your outcome could serve as a portfolio piece and a success case for your future applications.

Finally, we hope you find it enjoyable and suitable for your career goals within this module. We count on you to make this experience even better, starting now and for future iterations, so please do not hesitate to leave your feedback at any time.

Looking forward to making this journey incredibly successful,
Zezhi, Daniel, and the TA team.

Disclaimer: No Generative AI was used for the text above :)



II. Course Work Objective and Summary

Building a chatbot using Deep Learning techniques that will participate in a podcast. The podcast topic is at your choice but should address “AI literacy for everyone”, therefore the chatbot core LLM should be trained in adequate datasets.

A. Intended Learning Outcomes (ILO):

- Micro and macro applications of Deep Learning methods
- Applied experience in CNN for computer vision
- Dataset curating, handling, and training
- Large Language Models development and evaluation
- Evaluating models from methodological and users’ perspective

B. Main Challenges

- Students will need to discover how to look for the datasets themselves
- Students should identify and curate the target datasets
- The chosen datasets for images should present challenges to be handled (not converging quickly to a small loss error). Therefore, techniques for data handling should be applied (L1, L2, DataAug).

C. Group Size

- The groups should be formed by 4 people ideally. You can have groups of 3, but the workload would not be decreased because of that, as agreed during the lecture. However, in some special situations agreed with the teaching team, groups of 5 students will be allowed with an addition of an extra model implementation will be added for fair workload balance.

D. Checkpoints and evaluation

- There will be 2 checkpoints (not marked but mandatory)
 - The first one will specify and justify your decision choices of: audience, models, and approaches (week 5)
 - The second one will show us the datasets you will use, and the approach to train it, and the metrics used for both benchmarking and sustainability (week 7)
- You will be only marked over your deliverables of the **video podcast** and your **code** via github or hugging face. No final report is needed.



III. Breakdown of the project

A. Summary of the steps

- **Define your audience**
 - Children, high-school students, college students, general public, etc.
- **Choose your topic**
 - Robotics, sports, topics of your startup, AI literacy itself
 - Your topic does not need to be the same for the image and the dialogue models, but ideally it would be good for consistency's sake
- **Curate your dataset(s)**
 - Find and properly treat the best dataset for your chosen application
- **Implement your models**
 - Models for Image recognition and text generation (conversation)
- **Benchmark Evaluation**
 - Test your model in the benchmarks
- **User Evaluation**
 - Evaluate your model based on users' opinion
- **Record your video-presentation**
 - Pretty straightforward

B. Outcome Example

Audience: High-school students

Topic: AI for Wildlife Image Recognition

Dataset: Curated from Kaggle's "Animals-10" dataset (10 animal classes)

Model Implementation:

- **CNN** for image classification (PyTorch, transfer learning with ResNet-18) compared to your trained from scratch model.
- **Text Generation** chatbot to explain animal facts (using GPT-based model) and how a network is able to recognise the animals being showed.

Benchmark Evaluation:

- Test CNN on CIFAR-10 and Animals-10 benchmarks; report Top-1 accuracy

User Evaluation:

- Collect feedback from students on ease of use and educational value

Video Presentation:

- You show images to your system, and it shows the animals and tells characteristics about that given animal. Also, it explains briefly how it was able to recognize that given animal.



C. Models for Images

For the image recognition, you must implement two different types of architectures using transfer learning (not from scratch). They can serve for the same purpose (e.g. image classification, thus you compare their measures and explain the outcomes), or you can use the architecture for different purposes and combine them for a common goal (image generation, object identification,...).

1. In case the models are for the same purpose:

Please use **2 datasets per model** to show how the loss function and all the parameters of a classifier are dependent on the training dataset as well.

Note: Your 2 datasets can be the same dataset but with transformations. For instance, if you have a dataset A, by applying transformations techniques T to A, $T(A) = A'$, you have 2 different datasets (A and A'). Use them to compare the chosen metrics of your classifiers.

2. If you have the two models for two different tasks

If you have the two models for two different tasks, please consider comparing the models with a baseline for each one of them and reporting how these changes reflect on their metrics.

3. General comments

- You don't need to train from scratch, but you definitively need to improve their accuracy working on their training part.
- Original and more difficult ideas get higher marks.
- You get higher marks if your vision system is incorporated to your bot! But this is **not** compulsory.

Bonus: Ideas proposed by Copilot for how to use these networks are presented in chapter "Appendix

✦ Where CNNs Can Be Used in This Context" (click with Ctrl+click to go to this part) in the appendix of this document.

Hint: Show how the techniques used in the lectures helped your training (L1, L2, DataAug)...

4. Examples:

Image Classification (Baseline & Fine-tuned)

- **Fine-tuned model:** Use a pre-trained ResNet50 or MobileNetV2 and fine-tune on a custom dataset (e.g., flowers, fruits, traffic signs).



- **From scratch:** Build a small CNN (3–5 conv layers) and train it on the same dataset.
 - **Comparison:** Accuracy, F1-score, training time, overfitting behaviour.
 - **Explain outcomes:** Why transfer learning outperforms small CNN on limited data.
-

Object Detection + Classification

- **Fine-tuned model:** Use YOLOv8 or Faster R-CNN for object detection.
 - **From scratch:** Train a small CNN for classification of cropped objects.
 - **Integration:** Detection model finds objects → classification model labels them.
 - **Use case:** Detect and classify items in a kitchen or tools in a workshop.
-

Image Generation + Recognition

- **Fine-tuned model:** Use a pre-trained GAN or diffusion model for image generation (e.g., generate synthetic digits or faces).
 - **From scratch:** Train a CNN classifier on real + generated images.
 - **Goal:** Show how synthetic data improves performance.
-

Vision System for a Bot (Bonus Marks)

- **Scenario:** A robot that can recognize objects and respond.
- **Fine-tuned model:** Detect objects in the environment (YOLO or SSD).
- **From scratch:** Train a CNN to classify object states (e.g., open/closed door).
- **Integration:** Bot uses camera → detects object → decides action.

5. Full marks example for the vision system:

- Picking up a challenging dataset to deal with (a dataset that has no clear problem solution available, such as CIFAR and ImageNet have)
 - Ex: your own collected dataset or datasets to be recognized by literature to be challenging.



- Training a model to classify this dataset and compare it to the same model with a basic configuration.
- Training a different model to identify instances of the trained classes in real world images.
- Your chatbot has cameras and can make comments with the recognizable images.

IV. Dialogue models

For the dialogue generation part, you should consider 3 models:

- 1- **The dummy model:** A model trained from scratch in conservation datasets with very few epochs (max 10) for future comparison (not compulsory but extra marks)
- 2- **The core model:** A pre-trained model that you will fine-tune (this is where you should focus on)
- 3- **The state-of-the-art model:** An already trained or commercialized model (free version) that you can use in your code to serve as a baseline

A. The dummy model

Your dummy model is trained from scratch and will have a very poor performance due to be trained for few epochs. When you choose a benchmark for your core model, use it as well to evaluate your dumb model in the few epochs it was trained. Check how much the dummy increased during this N initial epochs and try to estimate in how many epochs it would get measures comparable with your core model.

This exercise is to make you (and your audience) aware of how hard it is to train models from scratch.

B. The core model

You **core model** should be trained using, **2 different datasets** at your choice (for instance one dataset for general Q&A and other datasets specific for your topic). You should achieve conversational levels of text generation and acceptable scores in the chosen benchmark.

A comparison of the core model finetuned in a different combination of dataset will give you extra marks. For instance, comparing the model finetuned in only one type of dataset with the model finetuned in the 2 of them.



C. The state-of-the-art model

This model is a consolidated model that you will use it to compare its benchmarks measures to the measures you obtained evaluating your dummy and core models.

D. Remarks

A. *Datasets for fine-tuning*

For instance, you can fine-tune your model with a conversational dataset and compare the results when you do not use this dataset for the training phase. Be creative! You can use datasets to personalise your chatbot with nice touches of your favourite writer. Can you imagine your favourite writer as a guest of your podcast?

B. *Benchmark evaluation*

Most of the available benchmarks come with some models already tested so you can compare your model's performance to other ones. Make sure you are also keeping track of the hallucinations of your bot.

C. *General remarks*

Make sure you know how to explain the steps you are taking to implement the models and any additional methods you use, such as RAGs, features pre-processing, data augmentation, etc.

Hint: you can play with the idea of having two different models. For instance, having two different bots that one will probably be smarter and the other and you explain why. Or even, one that classifies the images and gives only hints about the label of the picture so the other bot needs to guess what picture it is and maybe even recreate that image.

D. *If you are a group of 5 people:*

You will have to implement a model using LSTM. It could be a conversational model or another classifier that you find an application for it. In case of using as a conversational model, be aware that the accuracy will be much lower than transformers but explaining why it happens and the gains you can have with this implementation in terms of training and processing time can be definitively a plus for your podcast.

E. Extras

1. Features to your chatbot

You are encouraged to have extra features for your bot to increase user's enjoyment, such as text-to-speech (so your bot can speak by itself) and avatar faces for it.



2. Tests with Users

Between having your models ready and recording your podcast/video, ask for a group of people (around 5) to interact with your chatbot using the core and the state-of-art models (you can replace the state of art for a variation of the core trained with different datasets). Ask people to interact for a moment with your models and ask them questions related to their perception and enjoyment at the end. Analyse the data and compare the results with the ones obtained from benchmark evaluation phase. Inspirations for the user evaluation phase can be check at the “ Quantitative Methods to measure users enjoyment”, at the appendix of this document.

F. Summary of the conversational agent implementation

- Consider 3 models of conversational chatbot (Dummy, Core, state-of-the-art)
- Choosing datasets: Minimum 2 datasets in core model
- Evaluations:
 - Benchmarks
 - Tests with user
- Compare the results and drawn conclusions, for instance the ratio of user enjoyment versus the FLOPS it took to train the model.

V. Sustainability impact evaluation

Please propose a metric to evaluate the impact of your implementations on the environment. You can get inspiration from existing methods and build on top of them or be creative and come up with a completely new measurement (like counting the energy spent in Big Macs).

Take into consideration the following items because you will need to explain them later in your podcast:

- The designing process of your evaluation methodology
 - Inspirations, calculations, challenges
- Your module’s evaluation based on your methodology
 - Results, conclusions (users’ enjoyment vs energy consumption), pitfalls

VI. Structure of the podcast/ explainable video

Total: Not mandatory, but recommended between 20 and 30 minutes



1. Introduction

- Presenting the podcast, the members of the team, the objectives, etc
- Presenting your target audience, your constraints, and your reasoning for them
- Taster: Asking a humorous question to the model/chatbot about what Deep Learning is

2. Present your models

- Objective of each model (Image, dialogue generation, ...)
- Architecture, with parameters
- Datasets and training phase
- Benchmark Evaluations
- OBS: If you are using on-the-shelf models (for instance for TTS, avatar animation, etc) that you didn't develop, please use this space to briefly present them. Notice that paid features are not marked (meaning you cannot use models you have paid to use them as the core models).

3. Application demos

a. CNN for computer vision

Present the computer vision models you implemented as explained in section **Error! Reference source not found..** You can incorporate the vision networks in your bot (higher marks) or just show case them yourself, and ask the bot about what it thinks about the outcomes (lower marks).

b. Conversational models:

The model(s) should be able to answer questions and propose questions to keep the show going on. Please make comments and drawn conclusions linking the model's answers to the project decisions (for instance, the model's training phase and datasets to the type of answer. Models trained with dialogues datasets might have better dialogue creations while modules trained with educational datasets can offer more reflexive answers, leading to higher user self-reflexion). Regarding the two models, you should be able to explain their differences and analyse the results. For instance why finetuning in the dialogue datasets improves the bot's capability of having more enjoyable conversations.

Questions you should ask the bot to foster your podcast discussions:

- AI literacy:



- How does a computer learn?
- How can I explain my grandmother what is a Neural Network?
- Why do we need a lot of computational resources to process AI methods?
- Ethical considerations:
 - Will AI take over?
 - Can everyone benefit from it?
 - How to prevent AI to be “wrongly” used?

4. Sustainability discussion:

Please propose a metric to evaluate your full implementation’s impact on the environment. Use this section to present:

- The designing process of your evaluation methodology
 - Inspirations, calculations, challenges
- Your module’s evaluation based on your methodology
 - Results, conclusions (users’ enjoyment vs energy consumption), pitfalls

5. Wrap-up

- Summarize the key points,
- Inform which group member was responsible for each part
- Leave your email if you want people to reach out to you for more info (be very careful with that!)
- Ask for likes, subscriptions, money... (just kidding. Maybe the ‘money’ part could work though).

VII. Compulsory elements

Therefore, your coursework should present at least:

- **Two models for images:** Use the data treatment techniques covered during the lectures. If you implement models for different goals (e.g., one for image classification and another for image generation), simply acknowledge their evaluation metrics and compare them with existing models serving the same purposes.
- **Two models for conversational agents:** The dummy and the core.
- **If you are a group of five:** Implement a model using LSTM for a purpose of your choice. Your podcast may be longer than others due to this additional model presentation (for instance, it can extend up to 35 minutes).



- The **ethical and environmental evaluation and discussion** of your models
- Evidently, all these parts should be shown in your podcast/video, and it **must** follow the structure proposed in the previous sections to be accepted.

A. Minimal version

Therefore, for 50 marks you need:

- 2 models for image recognition (a CNN with explained techniques for training boost and a ResNet with the same) with different datasets than the ones showed in the lectures,
- 2 models for dialogue creation (dummy and core models) with the core model in conversational conditions and finetuned with 2 datasets,
- Discuss the ethical and environmental of your models,
- Your architecture and decisions are clearly explained in the video/podcast recorded in a Zoom or Teams meeting

The other 50 marks you achieve by being creative, implementing the different models for the image part, checking different fine tunings for the dialogue models, and doing the TTS and user evaluation part.

VIII. Limitations:

- Do NOT use paid features, paid resources, or paid models for implementations (one model for images and one of the conversational models). They may be used as part of the comparison models, but only as an extra. These should be acknowledged in the podcast.
- Use resources that do not exceed the maximum provided by the module (i.e., stay within the resource ceiling).

IX. Code delivery

You should upload on Moodle your links to the podcast and to your code repository.



X. Appendix

A. Where CNNs Can Be Used in This Context

1. Generating or Interpreting Visual Podcast Material

- **Context:** Podcasts often include **cover art**, social media **promo images**, or **illustrations** to support episodes.
- **Use of CNNs:** Students could use CNNs to **classify, tag, or generate captions** for AI-related images (e.g., robot images, neural network diagrams), and have the chatbot describe them.

2. Multimodal Chatbot Extension

- **Context:** Make the chatbot "see" and "talk".
- **Use of CNNs:** Integrate CNNs as an image understanding module—e.g., the user uploads an image, and the chatbot explains what it is in the podcast (like GPT-4V or CLIP).
- Students could simulate this using:
 - **Pretrained CNNs** (e.g., ResNet) to extract features
 - Then use **text generation models** to describe the image

3. Teaching AI Literacy Through CNN Demos

- **Context:** The chatbot might “explain” deep learning concepts to the audience.
- **Use of CNNs:** Students can train a simple CNN on MNIST or CIFAR-10 and have the chatbot **describe what a CNN is**, how it works, and show its outputs.
- The bot could respond to prompts like:

“How does a computer recognize a cat in an image?”

4. Image-to-Text as Input to Chatbot

- **Context:** A user might send a sketch or diagram related to AI.
- **Use of CNNs:** The CNN can process the image and generate a description (using a model like **Show and Tell**), which the chatbot uses to reply.

5. Fun/Easter Egg Features

- Students could build optional features:



- Chatbot plays a CNN “guess the image” game
- The chatbot creates AI-generated image quizzes

✅ Summary of Pedagogical Value

CNN Use Case

Educational Benefit

Explaining CNNs in the podcast Teaches core vision concepts

Using CNNs for image input Multimodal AI understanding

Captioning or tagging images Real-world AI application (e.g., accessibility)

Generating podcast visuals Integrates creativity with technical work

B. Quantitative Methods to measure users enjoyment

1. User Satisfaction Surveys

- Ask users to rate their experience (e.g., 1–5 stars or Likert scale).
- Include questions like: “Did you enjoy this interaction?” or “Was this chatbot helpful and engaging?”

2. Engagement Metrics

- **Session length:** Longer sessions may indicate enjoyment or interest.
- **Return rate:** Users who come back are likely enjoying the experience.
- **Interaction depth:** Number of messages exchanged or features used.

3. Sentiment Analysis

- Use NLP to analyze user messages for positive, neutral, or negative sentiment.
- Track changes in sentiment across sessions.

4. Task Success Rate



- If the chatbot helps users complete tasks (e.g., booking, learning), successful completions can correlate with satisfaction.

Example of Survey:



Chatbot Satisfaction Survey

- 1. How enjoyable was your interaction with the chatbot today?**
 - ☐ Very enjoyable
 - ☐ Enjoyable
 - ☐ Neutral
 - ☐ Not very enjoyable
 - ☐ Not enjoyable at all
- 2. Did the chatbot understand your questions and respond appropriately?**
 - ☐ Always
 - ☐ Most of the time
 - ☐ Sometimes
 - ☐ Rarely
 - ☐ Never
- 3. How engaging did you find the chatbot's tone and personality?**
 - ☐ Very engaging
 - ☐ Somewhat engaging
 - ☐ Neutral
 - ☐ Not very engaging
 - ☐ Not engaging at all
- 4. Would you choose to interact with this chatbot again in the future?**
 - ☐ Definitely
 - ☐ Probably
 - ☐ Not sure
 - ☐ Probably not
 - ☐ Definitely not
- 5. What could be improved to make your experience more enjoyable?**
(Open-ended response)

C. Datasets suggestion

Training or fine-tuning LLMs for **AI literacy** involves sourcing datasets that cover:



- Foundational AI concepts
- Ethics and societal impact
- Explainable AI (XAI)
- AI misconceptions
- Everyday applications of AI
- Human-AI interaction and literacy dialogue

Here's a categorized list of good datasets and sources, depending on your goals (chatbot, tutoring system, podcast co-host, etc.):

1. Educational & Scientific Datasets (Intro AI Concepts)

Dataset	Description	Use
AI2 Science Questions (Elementary–High School)	Science questions covering general knowledge, some AI-relevant	For Q&A or quizzes
OpenBookQA	Elementary-level science with explanations	Practice teaching factual reasoning
Cosmos QA	Reading comprehension with commonsense reasoning	For contextual conversation and reasoning
ExplainLikeImFive (Reddit)	ELI5 dataset from Reddit	To train models to simplify complex ideas
SciQ	Multiple-choice science questions with explanations	AI concepts + explainability tasks

2. AI Literacy & Public Understanding Datasets

Dataset	Description	Use
CommonSenseQA / SocialIQa	Commonsense questions for grounded reasoning	Helps explain AI reasoning to non-experts



Dataset	Description	Use
Wikipedia / Simple Wikipedia	Raw or simplified articles on AI and ML	Base for factual, explainable chatbot dialogue
Papers with Code + Abstracts	AI papers with easy summaries	Grounded AI literacy for academic users
AI Narratives (Oxford/Alan Turing Institute)	Human perceptions of AI in stories, media	Great for discussion about ethics and hype

3. Dialogue Datasets (for Chatbots)

Dataset	Description	Use
ConvAI2 / DailyDialog	Human-human conversations with emotions, logic	Teaching dialogue tone and flow
PersonaChat	Chat with personality; useful for tailoring LLMs to "AI educator" role	Add personality to the chatbot
Stanford Human-AI Interaction Dataset	Dialogue between humans and AI agents	Training for tutoring-style conversations

4. Educational Dialogues & Tutoring Corpora

Dataset	Description	Use
Coached Conversational Preference Elicitation (CCPE)	Google's dataset for AI tutoring systems	Good for modeling student-friendly LLMs
Tutoring Conversations Corpus	Dialogue between tutors and students	Model pedagogical responses and correction
Khan Academy Transcripts (via API scraping)	Explains educational topics including logic, statistics, ML	Train explanations and scaffolding techniques

 5. Ethics, Safety, and Societal Datasets

Dataset	Description	Use
ETHOS	Annotated dataset of ethical statements	For building LLMs that can reason ethically
Civil Comments	Identify respectful vs harmful content	Teach LLMs to explain AI risks and bias clearly
AllenAI Misconceptions Dataset	Contains common misconceptions in science	Train bot to clarify misunderstandings in AI literacy

 Tips for Customizing LLMs for AI Literacy

- **Use Retrieval-Augmented Generation (RAG):** Combine LLM with a curated knowledge base (e.g., distilled AI concepts).
- **Finetune on Q&A pairs + dialogue** for clarity and user adaptation.
- **Mix levels:** Include both expert and simplified content (ELI5 + PapersWithCode).

 Example Prompt Dataset You Could Create

Prompt	Response
“What is machine learning in simple terms?”	“It’s a way for computers to learn from data instead of being told what to do explicitly.”
“Can AI be conscious?”	“Current AI models do not have awareness or consciousness—they simulate responses based on data.”

Create a small, tailored **QA + dialogue** dataset like this using course materials or transcripts of podcasts/lectures.